

Carbon deposition in steam methane reforming over a Ni-based catalyst: Experimental and thermodynamic analysis

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ABSTRACT

The experimental investigation of the steam methane reforming process over an industrial Ni-based catalyst was presented. The set of experiments was performed in order to comprehend the effect of the carbon deposition on the methane conversion and pressure drop in a reformer. Various operating conditions such as temperature (600 °C and 800 °C), steam-to-methane ratio (0.5, 1.0, 2.0) and pressure were tested in the experiments. The thermodynamic analysis was accomplished to calculate the equilibrium carbon formation zones for various operating conditions and the experimental results were compared with the results of thermodynamic analysis. The experiments revealed that the methane conversion close to equilibrium is at a residence time of about $5 \text{ kg}_{\text{cat}} \cdot \text{s} / \text{mol}_{\text{CH}_4}$. The methane conversion as a function of the time on stream was experimentally determined. The maximum decrease in the methane conversion was observed for the steam-to-methane ratio (β) of 0.5. For $\beta = 2.0$ and $\beta = 1.0$, the decrease in the methane conversion is minimal. The reforming efficiency and mass of deposited carbon were determined for all investigated operation parameters. When the steam-to-methane ratio is greater than 1, the rate of carbon deposition has an almost linear dependence versus time on stream. For $\beta = 2$ and $T = 800 \text{ °C}$, the carbon deposition rate is approximately 0.12 g/h; for $\beta = 2$ and $T = 600 \text{ °C}$ - 0.21 g/h, for $\beta = 1$ and $T = 800 \text{ °C}$ - 0.29 g/h, for $\beta = 1$ and $T = 600 \text{ °C}$ - 1.02 g/h.

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1. Introduction

The methane steam reforming process is extensively used in various industries: chemical industry, energy, transport, etc. Steam methane reforming (SMR) is the main chemical reaction for large-scale production of hydrogen and synthesis gas in the chemical industry [1,2]. It is worth noting that methane steam reforming is the cheapest way of producing hydrogen [3]. In energy engineering, SMR is used in the thermochemical waste-heat recovery systems in order to enhance the energy efficiency of industrial furnaces [4–6], gas turbines [7], internal combustion engines [8], etc.

The SMR process is carried out, as a rule, in the presence of a catalyst at a temperature of more than 700–800 °C [9]. Various catalysts are used to enhance the efficiency of the SMR process: ruthenium (Ru) [10], rhodium (Rh) [11], palladium (Pd) [12], iridium (Ir) [13], and platinum (Pt) [14]. However, at an industrial scale, nickel-based catalysts are extensively used thanks to low cost and high efficiency [15,16]. Nickel catalysts are made in the form of

the particles of the various shapes: spheres, cylinders, Raschig rings, etc [17]. Kartik and Buwa found that not only operational parameters, such as pressure, temperature and composition of the initial reaction mixture, influence the SMR process, but also the shape of the catalyst [18].

When the SMR process takes place, in addition to the methane steam reforming reaction, many diverse reactions take place in the reformer [9,19]. Among these reactions, it is imperative to highlight the reactions leading to carbon formation. Such carbon can be deposited on the surface of the catalyst, thereby reducing the efficiency of the SMR process [20]. The studies concerning carbon deposition in the SMR process have been conducted by different scientists. Three main methods for studying carbon deposition for the SMR process can be distinguished: experimental research, thermodynamic analysis, and numerical modeling.

Bernardo et al. [21] have experimentally investigated the steam methane reforming process over the silica-supported Ni–Cu (nickel-copper) catalysts in the perspective of carbon deposition. The authors concluded that the adding steam to reaction mixture reduces the carbon formation and deposition in the temperature range before 800K. Also, they established that the catalyst

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composition has a significant effect on carbon formation. Silvester and co-authors [22] investigated the steam methane reforming process over the Ni-based catalysts for the chemical looping systems. The authors determined that the carbon deposition on the catalysts decreases monotonically with cycles. However, the decrease in the carbon deposition was much faster during the first 5 cycles and the carbon deposition continued to decrease slowly in subsequent cycles. Moreover, the authors reported that the composition of the catalysts is also playing a significant role in the amount of carbon deposition.

Nematollahi et al. [23] performed a thermodynamic analysis of different methane reforming processes in order to understand solid carbon formation. The authors reached to the conclusion that the carbon deposition decreases with increasing pressure in the methane reformer. For that reason, the industrial-scale steam methane reforming process is occurring at the pressure above 10 bar. Annesini and co-workers [24] presented the results of thermodynamic analysis based on the elemental composition to comprehend the carbon formation in the steam reforming process. They investigated the temperature ranges for the favourable conditions for carbon formation and deposition. The authors determined that in the temperature range of above 1000K, the carbon formation for the reaction mixture with steam-to-methane ratio above 1 is negligible.

Behnam and Dixon conducted the numerical investigation of the carbon deposition process [25] in the packed bed filled with the Ni-based catalysts. The results showed that a low steam-to-carbon ratio may cause local carbon formation at high temperature (about 900K) on the surface of the catalyst particle. The risk of carbon formation was highest at the surface hot spots and inside the catalyst particles where the methane cracking reaction rate exceeded those of the gasification reactions.

The carbon formation and deposition in the steam methane reforming process depends on various factors. These factors can be divided into two groups. The first group is operational parameters such as temperature, pressure, steam-to-methane ratio, residence time. The second group is reformer characteristics such as catalyst type and shape, design of the reformer, and its elements. Christensen et al. investigated an effect of supports and Ni crystal size on carbon formation and sintering during steam methane reforming [26]. The authors used conventional $\text{NiO}/\alpha\text{-Al}_2\text{O}_3$ catalysts at 823 K and a total pressure of 20 bar for various steam-to-methane ratios. They found that the higher resistance to carbon formation could be due to a higher saturation concentration of carbon in the smaller nickel crystals. Kumar and co-workers [27] reported the results of the creation of the new metal-support catalysts for the methane tri-reforming process with high coke resistivity. The authors concluded that the strength of metal-support interaction has a notable effect on the activity and stability of the Ni-based catalysts. In addition, the suitability of either strong or weak metal-support interaction is subject to catalyst composition.

To understand the effects of catalyst particle size on steam methane reforming activity over Ni-based catalyst, Wang et al. performed experimental and theoretical analysis [28]. The authors found that the small size of the Ni-based catalyst shows higher steam methane reforming activity than large ones. The authors observed that the carbon deposition phenomenon appeared on Ni(100). These results previously were found Schouten et al. [29]. In addition, Eizenberg and Blakely observed monolayer graphite formation on Ni-based catalyst surface during the experimental investigation of the steam methane reforming process [30].

In addition, the carbon deposition on the surface of the catalyst leads to an increase in the pressure loss in the reformer. Typically, a reformer is presented as a packed bed filled with the catalysts of various shapes [31]. Such reformers have a notable pressure loss,

but when carbon is deposited on the surface of the catalyst, pressure loss will increase. When carbon deposition on the surface of the catalyst takes place, the roughness of the catalyst particles is increased and as a consequence, the aerodynamic resistance of the packed bed increases. The effect of carbon deposition on reforming characteristics has been widely discussed in the scientific articles [32–34] but the works on the effect of carbon deposition on the pressure loss in the reformers has not been discovered by the author during a literature review. Nevertheless, the study of the effect of carbon deposition is an important task from a practical point of view. Additional roughness will lead to an increase in pressure loss and, as a consequence, to an increase in energy consumption for pumping the reaction mixture through a reformer.

Therefore, this investigation of the steam methane reforming process aims at assessing the effect of carbon deposition on the methane conversion and the pressure loss in the reformer. An experimental study of the steam methane reforming process and thermodynamic analysis were performed to achieve this aim. A wide variety of technological parameters (pressure, temperature, steam-to-methane ratio) as well as various catalysts (shape, composition, size) and reformers for the steam methane reforming process makes obtaining new experimental data an important task. Steam methane reforming is a widely investigated process. There are thousands of publications dealing with investigation SMR. This variety of publications is due to the effect of various parameters on the process. Steam reforming depends on temperature, pressure, composition, catalyst type, reformer type, catalyst composition and properties, and catalyst particle size and shape. In addition, the initial condition and the way of supplying heat to the reaction space also has a significant effect on the result. In our work, we used a commercial Ni-based catalyst and investigated practically intersected cases (reforming efficiency, pressure drop, the mass of deposited carbon, etc). We did not find similar studies with such initial data that we used. That is why we considered this research relevant and wrote this article.

2. Experimental and thermodynamics analyses

2.1. Experiment

The schematic diagram of an experimental setup for an experimental investigation of the steam methane reforming process is classical. The schematic diagram is shown in Fig. 1. The experiments were performed at the setup that can be virtually divided into three modules. The first module is to prepare the steam-methane mixture. The second module deals with implementing SMR and the third module is a measuring module that is meant for calculating and storing results.

The notations on the schematic diagram are as follows: 1 - a steam methane reformer filled the cylindrical catalysts; 2 - an electrical heater; 3 - a watt-hour meter; 4 - four thermocouples; 5 - a portable gas analyzer Gasboard 3100P; 6 - a laptop; 7 - the mass-flow meters; 8 - an electrical heat exchanger; 9 - an electrical steam generator; 10 - two manometers at the reformer inlet and reformer outlet.

The reformer is tubular. The outer diameter of the tube is 100 mm and the length of the tube is 800 mm. To stabilise flow before measurements, the region of 100 mm after inlet and 100 mm before outlet doesn't contain the catalyst. Therefore, the packed bed with the catalysts has a length of 600 mm. The electrical heater is heating the reaction mixture in the reformer up to 800–850 °C.

The synthesis gas composition is measured by Gasboard 3100P gas analyzer. This gas analyzer is connected to a laptop to store and calculate results. The steam generator PG-LEK 1.1 is used to produce superheated steam. High purity water (H_2O) is used in the steam

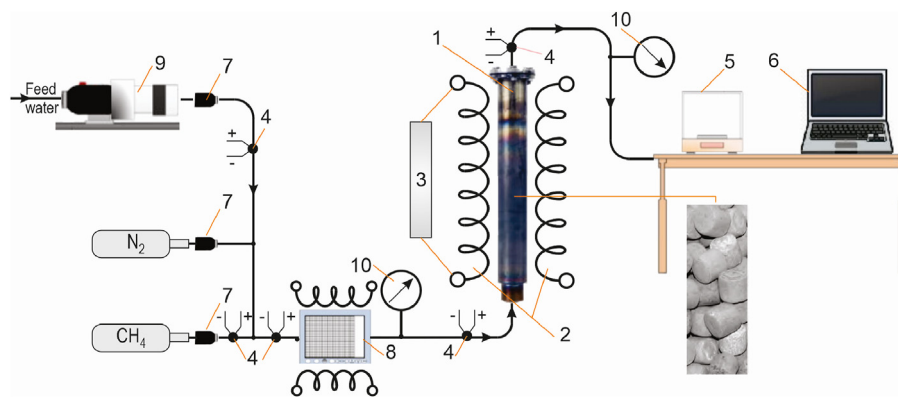


Fig. 1. Schematic view of experimental setup: 1 - a steam methane reformer filled the cylindrical catalysts; 2 - an electrical heater; 3 - a watt-hour meter; 4 - four thermocouples; 5 - a portable gas analyzer Gasboard 3100P; 6 - a laptop; 7 - the mass-flow meters; 8 - an electrical heat exchanger; 9 - an electrical steam generator; 10 - two manometers at the reformer inlet and reformer outlet.

generator. Methane N40 with 99.99% purity is used for the reforming process. The methane and nitrogen were obtained from the cylinders. Nitrogen is used as neutral gas for catalyst preheating. The flows of methane are measured by rotameters and the flow of steam is controlled by the mass-flow meters. Methane and steam were mixed to obtain the homogeneous mixture and this mixture was heated by an electrical preheater.

A photograph of a pilot setup and the main elements of the pilot setup are presented in Fig. 2. The notations in Fig. 2 coincide with the figure caption to Fig. 1.

The packed bed of the steam methane reformer is filled with the cylindrical $\text{NiO}-\alpha\text{Al}_2\text{O}_3$ catalyst. The composition of $\text{Ni}-\text{Al}_2\text{O}_3$ catalyst corresponds to an extensively used industrial catalyst: $\text{NiO} = 14.5\%$; $\text{SiO}_2 = 0.2\%$; support $\text{CaO}-\text{MgO}-\text{La}_2\text{O}_3-\alpha\text{Al}_2\text{O}_3$ [35]. The packed beds were obtained by random filling of the catalyst particles. The geometrical characteristics of the catalyst particle include: diameter – 15 mm, height – 15 mm.

In this study, the operation conditions such as temperature, inlet steam-to-methane ratio and residence time were changed to comprehend the effect of carbon deposition on reforming and flow

characteristics. The operation conditions were set as following: the temperature range is $600\text{--}800\text{ }^\circ\text{C}$; $\text{H}_2\text{O}/\text{CH}_4$ ratios are 0.5, 1 and 2; the pressure is 5 bar. The residence time (residence time i.e. catalyst pellet weight divided by the feed rate of the methane expressed as $\text{kg}_{\text{cat}} \cdot \text{s}/\text{mol}$ shows the mass of catalyst bed per mole flow-rate on methane inlet [36]) ranges from 2 to 6 $\text{kg}_{\text{cat}} \cdot \text{s}/\text{mol}$. The residence time was varied due to a change in the flow rate of the steam/methane mixture.

A set of experiments was performed. The fresh catalysts were used for each new experiment. The total number of catalyst particles in each experiment was constant and equal to 250 pieces. The mass of the deposited carbon on the catalyst surface was calculated as follows:

$$M_{\text{carbon}} = M_{\text{cat}}^{\text{after}} - M_{\text{cat}}^{\text{before}} \quad (1)$$

where $M_{\text{cat}}^{\text{after}}$, $M_{\text{cat}}^{\text{before}}$ is mass of catalysts before and after experiments, respectively.

An electronic scale accurate to 0.01 g is used to measure a catalyst mass.

In the present study, a set of experiments was performed for various operational parameters: a temperature range from 600 to $800\text{ }^\circ\text{C}$, a pressure 3 bar, steam-to-methane ratio from 0.5 to 2, a residence time (residence time i.e. catalyst pellet weight divided by the feed rate of the methane expressed as $\text{kg}_{\text{cat}} \cdot \text{s}/\text{mol}$ shows mass of catalyst bed per mole flow-rate on methane inlet) from 1 to 8 $\text{kg}_{\text{cat}} \cdot \text{s}/\text{mol}$. The residence time was varied due to a change in the flow rate of the reaction mixture.

Before the experiment, the electrical furnace heats the catalyst to the predetermined temperature. When the catalyst is heated, the reaction space of the methane reformer is filled with nitrogen. Heating of the catalyst takes approximately 15–17 min. The fresh catalysts are used for each experiment. During this period, the temperature in the reformer stabilizes throughout the reaction space. In parallel with the heating of the catalyst, steam generation is regulated by means of flowmeters. The temperature of the steam/methane reaction mixture is regulated in the heat exchanger to a temperature of the catalyst in the reformer.

At the initial time, the system of steam generation is not connected to the reformer. After the reformer becomes warmed up and the steam generation is regulated, the steam generation system is connected to the reformer. With the help of an electric heater, the temperature in the reaction space during the experiments is regulated. Each experiment has a duration of 10–15 min. Fresh catalyst is used for each experiment. During the experiment, the

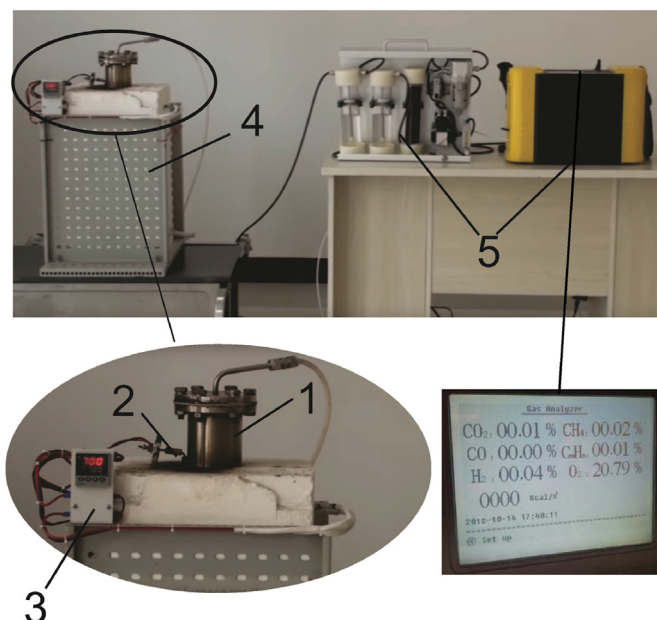


Fig. 2. The photographs of an experimental setup. The notation is the same as in Fig. 1.

composition of the reformed products and the heat consumption for the reforming reaction are fixed every minute.

2.2. Thermodynamics

In the steam methane reformer, various chemical reactions are occurring. The main reactions in the reformer are meant to produce H_2 (hydrogen), CO (carbon monoxide), CO_2 (carbon dioxide) and C (carbon). The reactions in the steam methane reforming process can be divided into two groups: methane oxidation and carbon formation. The main chemical reaction are shown in Table 1 [9]. Side chemical reactions may occur in the reformer, but these reactions are primarily a combination of the reactions presented in Table 1.

The favourable conditions for carbon formation can be determined by thermodynamic analysis. As it was shown in author's previous work [37] based on Gibbs free energy minimisation method, carbon formation can be favourable by other reactions: in the temperature range above $\sim 950K$ by reactions (4) and (5); in the temperature range below $\sim 950K$ by reactions (6) and (7) (the reaction number is from Table 1).

It is evident that among several possible reactions from Table 1, the most probable reaction is the reaction with the minimum of the Gibbs free energy. In this study, the FactSage database was used to determine the Gibbs free energy change [38]. The change in the Gibbs free energy as a function of temperature for the chemical reaction from Table 1 is presented in Fig. 3. The reactions in Fig. 3 are numbered in the same way as in Table 1.

To prevent the carbon formation in the reformer, the following conditions must be fulfilled:

- for temperature above $\sim 950K$:

$$\frac{P_{H_2}^2}{P_{CH_4}} > K_{eq,4}; \frac{P_{CO_2}}{P_{CO}^2} > K_{eq,5}; \quad (2)$$

- for temperature below $\sim 950K$:

$$\frac{P_{CO} \cdot P_{H_2}}{P_{H_2O}} < K_{eq,6}; \frac{P_{CO_2} \cdot P_{H_2}^2}{P_{H_2O}^2} < K_{eq,7}; \quad (3)$$

where P_j – is partial pressure of j th element; j -index for the equilibrium constants $K_{eq,i}$ is number of reaction in Table 1.

To determine the favourable conditions for the carbon formation, the following new variables are introduced:

- steam-to-methane ration can be expressed as:

Table 1
Possible chemical reactions in the reformer [9].

No.	Reaction	ΔH_{298} , (kJ/mol)
	Methane oxidation	
1	$CH_4 + H_2O \rightleftharpoons CO + 3H_2$	206.4
2	$CO + H_2O \rightleftharpoons CO_2 + H_2$	-41.1
3	$CH_4 + CO_2 \rightleftharpoons 2H_2 + 2CO$	247.3
	Carbon formation	
4	$CH_4 \rightleftharpoons C + 2H_2$	74.8
5	$2CO \rightleftharpoons C + CO_2$	-173.3
6	$CO + H_2 \rightleftharpoons C + H_2O$	-90.13
7	$CO_2 + 2H_2 \rightleftharpoons C + 2H_2O$	-131.3

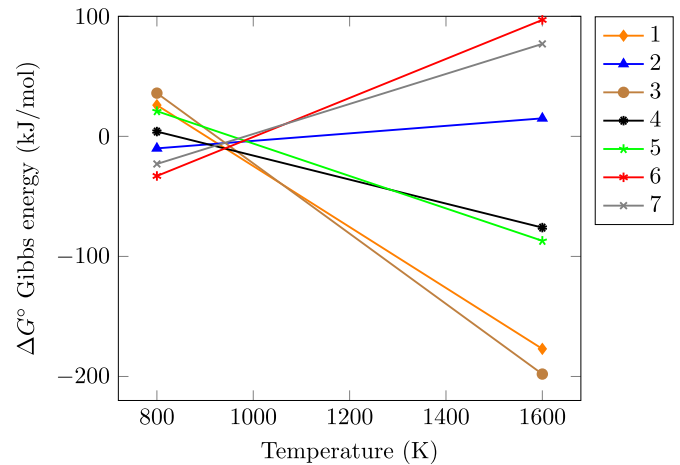


Fig. 3. The change of Gibbs free energy as a function of temperature: the number of each line corresponds to the number of reaction in Table 1.

$$\beta = \frac{\dot{n}_{H_2O,1}}{\dot{n}_{CH_4,1}}. \quad (4)$$

- the methane conversion (X_{CH_4}) and the methane conversion to carbon dioxide (X_{CO_2}):

$$X_{CH_4} = \frac{\dot{n}_{CH_4,1} - \dot{n}_{CH_4,2}}{\dot{n}_{CH_4,1}}; X_{CO_2} = \frac{\dot{n}_{CO_2,2}}{\dot{n}_{CH_4,1}}. \quad (5)$$

In these equations $\dot{n}_{ij}$ is the molar flow rate of i -element through inlet (subscript 1) and outlet (subscript 2), mol/s.

The carbon formation conditions is determined from in-equation:

$$\dot{n}_{CH_4,1} - \dot{n}_{CH_4,2} - \dot{n}_{CO,2} - \dot{n}_{CO_2,2} > 0 \quad (6)$$

To determine equilibrium molar flow rate of i -element at the reformer outlet, a system of equations with the material balance equations and the law mass equation must be solved. A detailed algorithm for solving the system of equations has been presented in the author's previous work [39].

3. Results and discussion

In this section, we presented the results of thermodynamic analysis and experimental results. The real efficiency of the steam methane reforming process depends on various parameters such as temperature, pressure, composition, catalyst type, reformer type, catalyst composition and properties, and catalyst particle size and shape, etc. Therefore the experimental results obtained in this study can be used for the development of new models or algorithms, and verification of the theoretical results (for example, numerical simulation or CFD-modeling).

3.1. Equilibrium conditions

In the scientific literature, there is adequate information that carbon formation can be regulated by the operating conditions such as a steam-to-methane ratio [40]. In this subsection, the

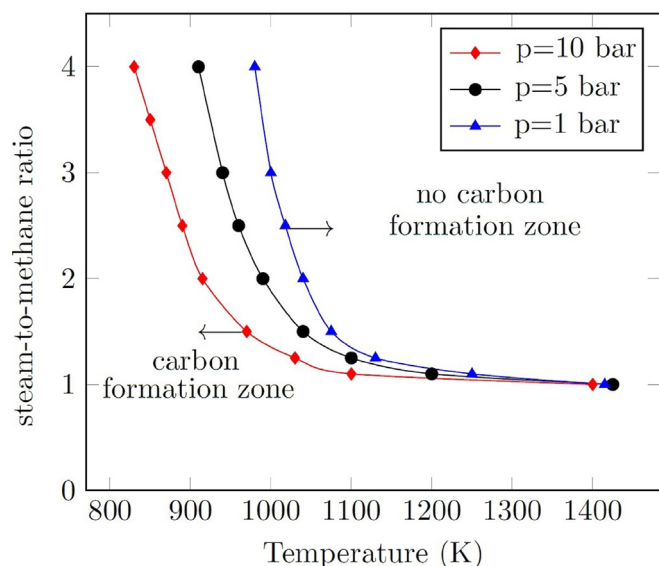


Fig. 4. The operational conditions for carbon formation in SMR.

steam-to-methane ratio was varied from 0.5 to 4.0.

Fig. 4 shows the carbon formation zones and no carbon formation zones for various operating conditions such as steam-to-methane ratio, temperature, and pressure. The area to the left of each curve is the area of carbon formation for each pressure. The area to the right of each curve is the no carbon formation zone for each pressure. Fig. 4 shows that for the steam-to-methane ratio less than 1 there is not enough oxidizer to oxidize and prevent the carbon formation. Therefore, for the steam-to-methane ratio less than 1, the carbon formation is observed for all operating conditions. Fig. 4 depicts that the carbon formation area decreases when pressure is decreasing. The effect of pressure on carbon deposition is an important issue for the life cycles of the reformers. Carbon formation suddenly increased by the decrease in pressure at all the range of the ratio of feedstock. This conclusion is correlated with the results obtained by Tabrizi et al. [41]. In addition, Fig. 4 shows that an increase in the steam-to-methane ratio in the reaction mixture reduces the temperature at which carbon formation occurs. This is also the reason that for the large-scale SMR, the steam-to-methane ratio greater than 2 is used for the steam methane

reforming process [1].

3.2. Methane conversion

To analyse the effect of the carbon deposition on the methane reforming, the effect of temperature on Gibbs free energy change was accessed. The carbon deposition on the catalyst surface reduces the activity of the catalyst. Therefore, the more carbon deposits on the catalyst surface, the lower the efficiency of the reforming process. Consequently, lower methane conversion is observed. The methane conversion is the ratio of reacted methane to the initial methane content in the reaction mixture. According to the definition, the methane conversion can be expressed by eq. (5).

The methane conversion is affected not only by the operating parameters, such as temperature, pressure and the inlet steam-to-methane ratio, but also by the flow rate of the reaction mixture, type, and shape of catalyst [9,18]. It is eminent that the equilibrium condition of the steam methane reforming reaction occurs with a moderately long contact of the reaction mixture in the reformer. The contact time of the reaction mixture in the reformer is called residence time. Residence time can be expressed as the mass of the catalyst per mol/s of inlet methane.

In this study, a series of experiments was conducted to determine the residence time at which the products of the SMR reaction reach a composition close to equilibrium. This is necessary to compare the experimental results with the results of thermodynamic analysis. The duration of each experiment in this series of experiments is about 15 min.

Fig. 5 shows the comparison of the experimental results with the results of the thermodynamic analysis. In this figure, the equilibrium methane conversion was obtained from the previous author's works [39]. The experiments are conducted for $\beta = 1$ and $\beta = 2$. By increasing the residence time, the methane conversion approaches to equilibrium value (dashed lines). Fig. 5 shows that the methane conversion close to equilibrium is achieved at a residence time of about $5 \text{ kg}_{\text{cat}} \cdot \text{s} / \text{mol}_{\text{CH}_4}$.

Fig. 6 shows the methane conversion as a function of the time on stream at residence time of about $5 \text{ kg}_{\text{cat}} \cdot \text{s} / \text{mol}_{\text{CH}_4}$. Measurements were made at the end of each time interval (time step): Fig. 6a every 3 min and Fig. 6b every 30 min. The experiments were performed at $T = 800^\circ\text{C}$ and $T = 600^\circ\text{C}$ for various steam-to-methane ratios: $\beta = 0.5$, $\beta = 1.0$, $\beta = 2.0$. Fig. 6a shows that at a temperature of 800°C for $\beta = 2.0$ and $\beta = 1.0$, the methane conversion does not change significantly. Therefore, carbon deposition on the surface in

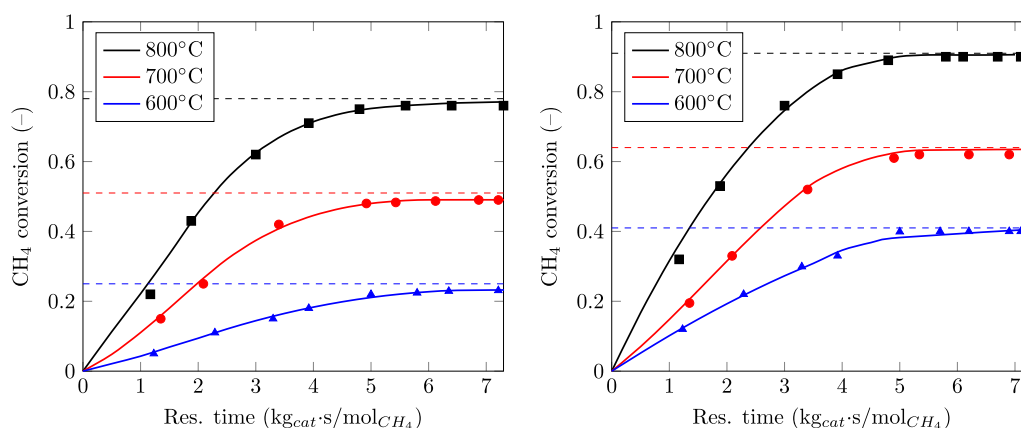


Fig. 5. Effect of temperature on CH_4 conversions at $p = 5$ bar for the various inlet gas compositions (a – $\beta = 1$; b – $\beta = 2$): dashed line – equilibrium condition; solid line – approx. curve; hollow point – experimental results.

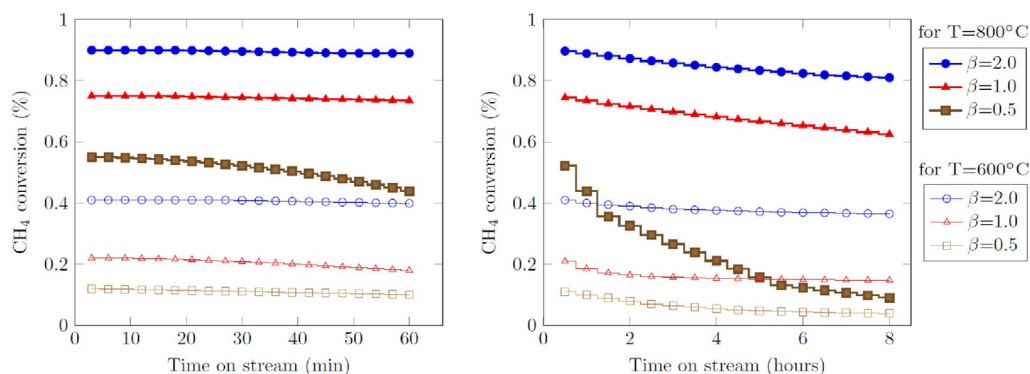


Fig. 6. CH₄ conversions vs. time on stream: a – time on stream in minutes; b – time on stream in hours; filled points – for T = 800 °C; hollow points – for T = 600 °C.

60 min is insignificant for these operational parameters. This can be explained by the fact that according to Fig. 3, at a temperature of 800 °C, the change in the Gibbs energy for the methane steam reforming reaction (1) is more than two times smaller than the change in the Gibbs energy for the methane cracking reaction (4) and the Boudouard reaction (5). In addition, in the temperature range above 800 °C, reactions (6) and (7) have the opposite direction, which leads to gasification of the formed carbon by reactions (4) and (5). However, if for conditions 800 °C and $\beta = 2.0$, carbon formation is thermodynamically unfavourable (according to Fig. 4) then for conditions 800 °C and $\beta = 1.0$, the system is in the carbon formation zone. Due to a decrease in catalyst activity due to carbon deposition, a noticeable decrease in the methane conversion is observed (Fig. 6a) at 800 °C and $\beta = 0.5$. The deposition of carbon on the catalyst surface reduces the reaction rate of steam reforming and, as a consequence, reduces the methane conversion.

At a temperature of 600 °C, all three investigated steam-to-methane ratios of $\beta = 2.0$, $\beta = 1.0$, $\beta = 0.5$ are thermodynamically in the carbon formation zone (Fig. 4). For T = 600 °C, methane is predominantly consumed by the methane steam reforming reaction (1) and methane cracking reaction (4). In this case, the change in the Gibbs energy for the steam reforming reaction is approximately equal to the change in the Gibbs energy for the

cracking reaction. However, the methane conversion decreases slightly. Thus, it can be assumed that the carbon formed by the cracking reaction (4) is consumed by the Boudouard reaction (5). With $\beta = 1.0$, the decrease in the methane conversion with time on stream becomes more noticeable. Within 60 min, the methane conversion decreases by more than 12%. For $\beta = 0.5$ at T = 600 °C, a minimal change in the methane conversion is observed. This can be explained by the fact that methane is predominantly consumed by the cracking reaction, and the carbon is partially consumed by another reaction from Table 1. In addition, carbon deposition on the surface of the catalyst slightly affects the rate of the cracking reaction, and the rate of methane reforming is negligible at T = 600 °C.

Fig. 6b clearly shows a trend towards a decrease in the methane conversion with an increase in the duration of experiments. The largest decrease in methane conversion is observed for $\beta = 0.5$ at T = 800 °C. Moreover, for $\beta = 2$ at T = 800 °C and T = 600 °C, the change in the methane conversion is minimal. This fact agrees well with the results of the thermodynamic analysis since the operating conditions of $\beta = 2$ and T = 800 °C are located near the boundary curve of carbon formation. It is worth mentioning that at $\beta = 0.5$ and T = 600 °C, the methane conversion does not change significantly in comparison to $\beta = 0.5$ and T = 800 °C. This can be

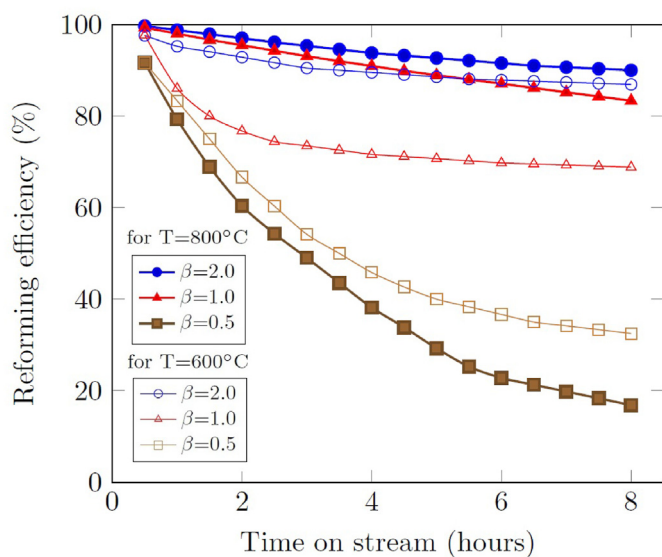


Fig. 7. Reforming efficiency vs. time on stream: filled points – for T = 800 °C; hollow points – for T = 600 °C.

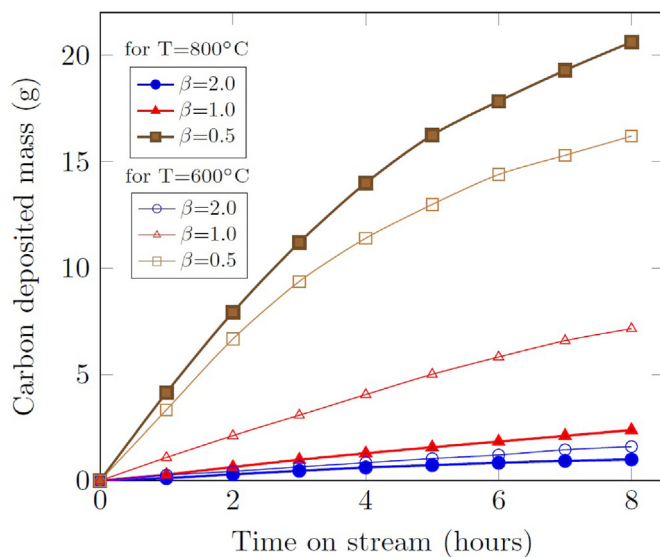


Fig. 8. Mass of deposited carbon vs. time on stream: filled points – for T = 800 °C; hollow points – for T = 600 °C.

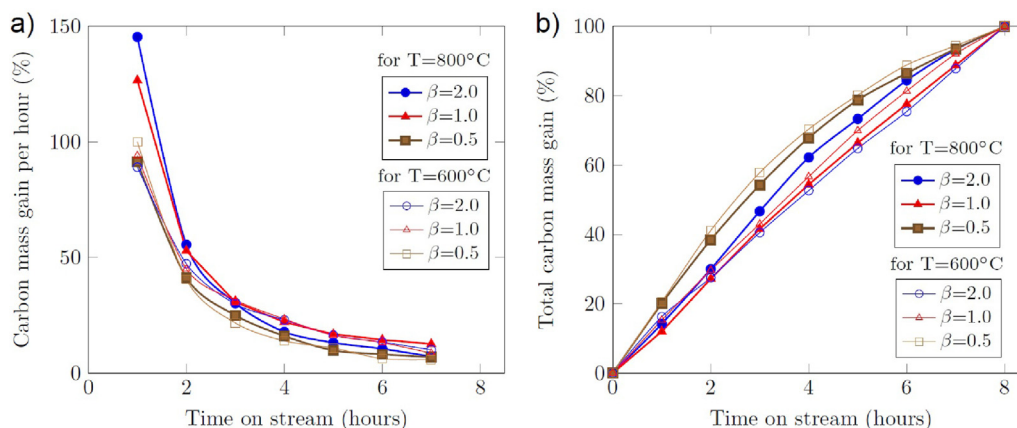


Fig. 9. Carbon mass gain as a function on time on stream (duration of experiment): a) carbon mass gain per hour; b) total carbon mass gain. Filled points – for $T = 800^\circ\text{C}$; hollow points – for $T = 600^\circ\text{C}$.

explained by the fact that at $T = 600^\circ\text{C}$, the behaviour of a methane cracking reaction (4) is most likely and the probability of a methane reforming reaction (1) is small, therefore, methane is predominantly consumed by reaction (4). In this case, the Ni-based catalyst mainly increases the rate of the reforming reaction (1) and it has a minimal effect on reaction (4). Therefore, carbon deposition slightly reduces the methane conversion at $T = 600^\circ\text{C}$ for $\beta = 0.5$. The finding results are in good correlation with other studies on effect of carbon deposition on reforming characteristics [9,42].

3.3. Carbon deposition

To understand the effect of carbon deposition on the reforming efficiency with time, the dependence in Fig. 7 was obtained based on experimental results from Fig. 6. The reforming efficiency was calculated as follows:

$$E_{\text{ref}} = \left(1 - \frac{X_{\text{CH}_4}^1 - X_{\text{CH}_4}^i}{X_{\text{CH}_4}^1} \right) \times 100\%, \quad (7)$$

where $X_{\text{CH}_4}^1$ – the methane reforming at the first time step (3 min after start of experiment); $X_{\text{CH}_4}^i$ – the methane reforming at the i th time step of experiment.

Fig. 7 shows that at the temperature of 800°C and with a steam-to-methane ratio of 2, methane reforming efficiency decreases slightly in comparison to the steam-to-methane ratio below the stoichiometric value ($\beta = 0.5$). For this reason, in large-scale industry, steam methane reforming is carried out at $\beta = 2$ and a temperature of more than 800°C . For condition $\beta = 2$ and $T = 600^\circ\text{C}$, a decrease in reforming efficiency is observed initially. Then there is a gradual decrease in efficiency of about 1% every hour.

To determine the mass of deposited carbon in the packed bed of

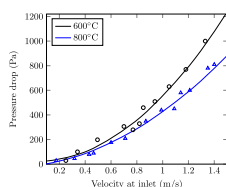


Fig. 10. Pressure drop vs. inlet velocity for various temperatures after 10 min of experiment: the hollow points – experiment; the lines – approximation curve.

the reformer, a series of experiments were performed. The duration of each experiment was 1 h. The number of catalyst particles in each experiment was constant (250 pieces). After each experiment, the catalysts were weighed and the mass of deposited carbon was determined by expression (1). Fig. 8 shows the dependence of the mass of deposited carbon on the time on stream.

Fig. 8 shows that when the steam-to-methane ratio is greater than 1, the rate of carbon deposition has almost linear dependence on time on stream. For $\beta = 2$ and $T = 800^\circ\text{C}$, the carbon deposition rate is approximately 0.12 g/h; for $\beta = 2$ and $T = 600^\circ\text{C}$ - 0.21 g/h, for $\beta = 1$ and $T = 800^\circ\text{C}$ - 0.29 g/h, for $\beta = 1$ and $T = 600^\circ\text{C}$ - 1.02 g/h. For $\beta = 0.5$, the carbon deposition rate increases notably. In addition, the carbon deposition rate is maximum initially. When the time on stream increases, the carbon deposition rate decreases. For $\beta = 0.5$ and $T = 800^\circ\text{C}$, for the second hour of the experiment, the carbon deposition rate is 3.82 g/h, while for the last hour of the experiment, the carbon deposition rate is 1.37 g/h. The slowdown in the rate of carbon deposition can be explained by a general decrease in the efficiency of the reforming process, including methane cracking reactions. When the efficiency of the reforming process is maximum (according to Fig. 7), the carbon deposition

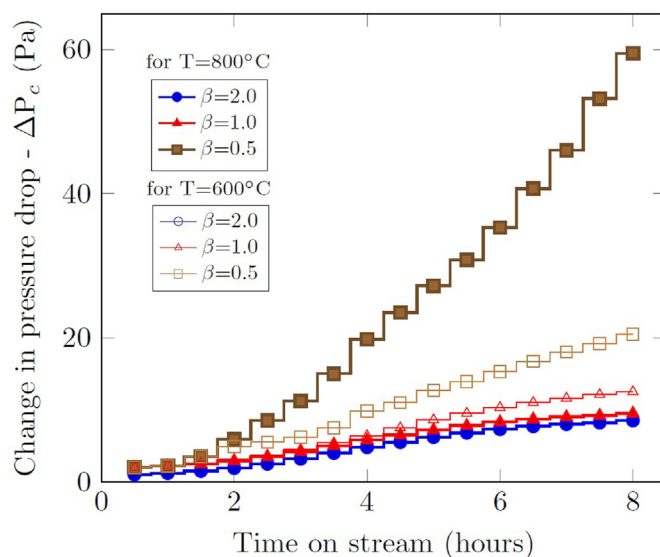


Fig. 11. Pressure drop vs. time on stream for various temperatures: filled points – for $T = 800^\circ\text{C}$; hollow points – for $T = 600^\circ\text{C}$.

rate is maximum.

To evaluate and compare the carbon deposition rate, the carbon mass gain for the steam methane reforming process at the residence time of about $5 \text{ kg}_{\text{cat}} \cdot \text{s/mol}_{\text{CH}_4}$ was determined. For this, the following two indicators were identified:

- carbon deposition gain per hour (%):

$$C_{\text{hour}} = \frac{M_{C_{i+1}} - M_{C_i}}{M_{C_i}} \times 100\% \quad (8)$$

- total carbon deposition mass gain (%):

$$C_{\text{total}} = \frac{\sum M_{C_i}}{M_{C_{\text{total}}}} \times 100\% \quad (9)$$

In these expressions, M_{C_i} is mass of deposited carbon for i th time step, $M_{C_{\text{total}}}$ is mass of deposited carbon after 8-h experiment (8 h is duration of experiment).

Fig. 9a shows the carbon deposition gain per hour and Fig. 9b shows the total carbon deposition mass gain. Fig. 9a shows that carbon deposition gain per hour has the maximum increase for all studied operational parameters during the initial steps. Also, it can be seen from Fig. 9a that the carbon deposition gain per hour for all the studied operational parameters has similar dynamics. This can be explained by the fact that despite the difference in the steam-to-methane ratio, carbon is predominantly formed by reactions (4) and (5) (Table 1), for which the steam concentration does not matter. Fig. 9b shows the carbon deposition rate for every hour. It can be seen from Fig. 9b that for $\beta = 1$ and $\beta = 2$, approximately equal mass of carbon deposits every hour. However, for $\beta = 0.5$, it is seen that during the initial time in the first 3 h, the mass of carbon deposits faster than that in the last 3 h of the experiment. This can be explained by the fact that deposited carbon in the reformer slows down the carbon formation reactions (according to Table 1).

3.4. Pressure drop

To evaluate the effect of deposited carbon on pressure drop, a series of experiments were performed. Fig. 10 shows the dependence of the pressure drop on the inlet velocity for various temperatures.

In Fig. 10, the pressure drop was measured during 10-min experiment. Therefore, it is expected that during the first 10 min, carbon deposition does not have a significant effect on the pressure drop. In addition, Fig. 10 shows the experimental results only for $\beta = 2.0$. This is due to the fact that the pressure drop for $\beta = 1$ and $\beta = 0.5$ almost completely coincides with the pressure drop for $\beta = 2.0$. Fig. 10 shows that the approximation curves for $T = 600^\circ\text{C}$ and $T = 800^\circ\text{C}$ have a characteristic quadratic dependence. As velocity increases, pressure drop increases faster. A smaller pressure drop for $T = 800^\circ\text{C}$ as compared to $T = 600^\circ\text{C}$ can be explained by the fact that according to the classical concept of aerodynamics, the pressure drop in a packed bed depends on the gas density [43]. By increasing temperature, the density of the reaction mixture decreases. In addition, at $T = 800^\circ\text{C}$, the reforming products contain mainly gases with a low density.

With an increase in the duration of the experiment, the effect of carbon deposition becomes noticeable. Fig. 11 shows the dependence of the change in pressure drop due to carbon deposition for an initial velocity of 1 m/s .

In Fig. 11, the change in pressure drop is defined as the pressure difference:

$$\Delta P_c = P_{\text{out}}^t - P_{\text{out}}^0 \quad (10)$$

where P_{out}^t - outlet pressure for i -th time step, Pa; P_{out}^0 - outlet pressure after 10-min experiment, Pa.

Fig. 11 shows that for $\beta = 1$ and $\beta = 2$, the effect of carbon deposition is minimal and is only a few percent. However, for $\beta = 0.5$ at a temperature of 800°C , the change in pressure drop noticeably increases with time. Therefore, after an 8-h experiment, the pressure drop increased by about 15% in comparison to the pressure drop for the 10-min experiment. For such operating conditions ($T = 800^\circ\text{C}$, $\beta = 0.5$), the maximum carbon deposition was just observed.

4. Conclusion

In this paper, the experimental and thermodynamic study of the steam methane reforming process has been conducted to comprehend the carbon deposition in the reformer. Based on the thermodynamic analysis the equilibrium conditions for the carbon deposition were determined. The increase in pressure and the steam-to-methane ratio in the reaction mixture leads to decrease in the carbon formation zone. The experiments show that by increasing the residence time, the methane conversion approaches to equilibrium value and the methane conversion close to equilibrium is achieved at a residence time of about $5 \text{ kg}_{\text{cat}} \cdot \text{s/mol}_{\text{CH}_4}$. The methane conversion as a function of time on stream is determined for various time steps: every 3 min and every 30 min. The maximum decrease in the methane conversion is observed for the steam-to-methane ration (β) of 0.5. For $\beta = 2.0$ and $\beta = 1.0$ the decrease in the methane conversion is minimal. The reforming efficiency and mass of deposited carbon are determined for all investigated operation parameters. When the steam-to-methane ratio is greater than 1, the rate of carbon deposition has an almost linear dependence on time on stream. For $\beta = 2$ and $T = 800^\circ\text{C}$, the carbon deposition rate is approximately 0.12 g/h ; for $\beta = 2$ and $T = 600^\circ\text{C}$ - 0.21 g/h , for $\beta = 1$ and $T = 800^\circ\text{C}$ - 0.29 g/h , for $\beta = 1$ and $T = 600^\circ\text{C}$ - 1.02 g/h . The carbon deposition gain per hour and the total carbon deposition mass gain are calculated based on the experimental results. It is observed that the carbon deposition gain per hour for all the studied operational parameters has similar dynamics. For $\beta = 1$ and $\beta = 2$ the carbon deposition rate for every hour has approximately equal mass of carbon depositing every hour. However, for $\beta = 0.5$, it is seen that during the initial time in the first 3 h, the mass of carbon deposits faster than in the last 3 h of the experiment.

Credit author statement

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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